Monish V

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Professional Summary

Computer Science undergraduate with a strong foundation in **C++, Data Structures & Algorithms, and problem-solving**. Hands-on experience building and deploying full-stack applications using **React.js, Node.js, Express.js, and MySQL**, alongside developing C++ systems software and Python-based backend services.

Education

Acharya Institute of Technology — Expected 2027
B.E. Computer Science & Engineering — *Bengaluru, India*

Projects

CacheForge — Thread-Safe In-Memory Cache Engine | *C++17, STL, CMake* 2026

- Engineered a thread-safe key-value cache in C++17 using **std::unordered_map** and an intrusive doubly linked list for O(1) LRU eviction.
- Benchmarked 100K operations across 5 runs on a Ryzen 5 3500U (Release build), achieving median throughput of **2.86M inserts/sec, 12.75M reads/sec, and 10.37M mixed ops/sec**.
- Added **TTL expiration, configurable capacity, statistics, and file persistence**; packaged as an installable CMake library with automated tests. [GitHub]

SkillTree — Student Skill Tracking Platform | *React.js, Node.js, Express.js, MySQL, JWT* 2026

- Developed and deployed a full-stack placement-prep platform combining resume analysis, skill tracking, career roadmaps, and interview practice.
- Implemented **JWT authentication, password reset, PDF resume parsing, and automated skill extraction** through RESTful Node.js/Express.js services backed by MySQL.
- Deployed the frontend on Vercel. [Live] [GitHub]

ReleasePilot — AI Release Readiness Analyzer | *React.js, Node.js, Express.js, IBM Bob 2.0* 2026

- Built an AI-assisted tool that analyzes repositories for release readiness across code quality, testing, documentation, and configuration; validated on **2 real repositories**.
- Implemented specialized analysis agents for **code-risk detection, test analysis, and documentation/configuration checks**, producing evidence-backed findings including 18 persisted findings in a documented analysis run.
- Implemented **asynchronous analysis, report persistence, and real-time status tracking** with a React.js frontend. [Live] [GitHub]

RoadSense — Traffic Risk Detection System | *Python, FastAPI, Scikit-learn, React* 2026

- Trained a **Random Forest classifier** (100 estimators) on **50,000 records with 8 traffic, road, environmental, and driver features**, reaching **99.52% accuracy** on a 10,000-record held-out test set (80:20 split) across Low/Medium/High risk classes.
- Served predictions through a **FastAPI REST service** with JWT authentication and confidence scoring.
- Integrated browser geolocation with **Leaflet/OpenStreetMap** for location-based predictions, history, and regional risk patterns. [GitHub]

Achievements & Activities

- Presented a research paper on **AI-Based Traffic Risk Detection using Machine Learning** at a conference.
- Built **ReleasePilot** at the **lablab.ai x IBM Bob 2.0 Hackathon**, an AI-assisted software release readiness analyzer.
- Participated in the **Scaler Hackathon** and **Binary Battle** (competitive programming event).

Technical Skills

Languages: C++, Python, SQL, JavaScript
Web & APIs: React.js, Node.js, Express.js, FastAPI, REST, JWT, HTML, CSS, Bootstrap
Databases: MySQL, FAISS (vector search)
Core CS: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Multithreading
AI/ML: Scikit-learn, Pandas, NumPy, SentenceTransformers, RAG, Groq API, Model Evaluation
Tools: Git, GitHub, CMake, Streamlit, Postman, Leaflet, VS Code